

ALMA MATER STUDIORUM – UNIVERSITY OF BOLOGNA

SCHOOL OF ENGINEERING AND ARCHITECTURE

DEPARTMENT OF INDUSTRIAL ENGINEERING-DIN

SECOND CYCLE DEGREE IN MECHANICAL ENGINEERING

GROUP PROJECT

in

Chassis and Body Design and Manufacturing

PROFESSOR:

Massimiliano De Agostinis

CANDIDATES:

Coliva Lorenzo
Fratini Matteo
Gabrielli Luca
Marcaccini Marco
Novi Martina
Suozzi Chiara

Abstract

The project's aim is to develop a handlebar bracket for a sport motorcycle. To accomplish the task, are given boundary conditions such as maximum spatial boundaries (because of the presence of shock absorber, steering plate, and handlebar), maximum static loads and fatigue loads (and so the expected lifetime of the component), the material and the maximum fasteners class. The procedure of design of the bracket has been guided by the topology optimization, followed by further investigations of static analysis, fatigue analysis and others.

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1 Assignment

The goal of the project was to design an adjustable handlebar bracket for a super sport motorbike, as shown in the Picture below.



Figure 1.1: Handlebar bracket of a sport motorbike

1.1 Spatial Boundaries

The spatial boundaries were assigned through a 3D CAD model (Picture 1.1) with technical drawing specifications (Picture 1.2 (a)) supporting the model itself. There are 4 spatial conditions:

- A lower boundary surface, only acting on the intersection with the stanchion limiting the lower space available.
- A cylindric boundary surface, concentric with respect to the stanchion, limiting the inner space of the bike.
- A forward boundary surface with respect to the stanchion, limiting the front space available.
- The lower boundary of the steering plate, limiting the upper space available.

1.2 Load Condition Assignment

The handlebar bracket will be subjected to both a static and dynamic load. According to the frame of reference shown in Picture 1.2(b), the assignment on the static load conditions consists in an ultimate vertical load, named F_Z , of 150 N and an ultimate transverse load, named F_X , of 400 N that the structure must withstand with no permanent deformations, and with admissible elastic deformation. The two loading components must be applied as shown in Picture 1.2(b).

Referring to fatigue, the bracket needs to be verified at 10^5 cycles. The fatigue cycle assigned to our group consists of a constant magnitude of F_Z , equal to -60 N and an alternated F_X of ± 300 N.

Table 1.1: Load values

F_Z [N]	F_X [N]	F_{Z_max} [N]	F_{X_max} [N]
60	+/-300	150	400

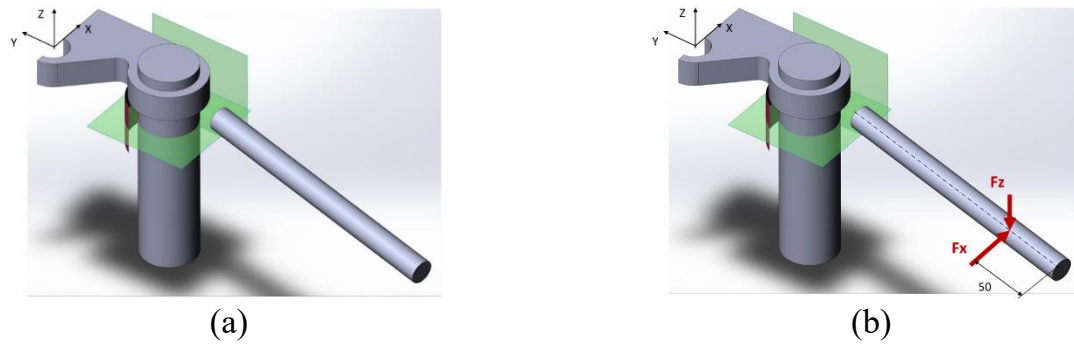


Figure 1.2: (a) Boundary Conditions; (b) Load configuration

1.3 Joints configuration

The bracket will be connected to the stanchion of the shock absorber, and to the handlebar, through friction connections. To be adjustable by 20° along the axis of the stanchion, and by 90° along the axis of the handlebar, we chose to design two clamps that will interface with the already citated bodies. Each clamp manages to create the correct contact pressure through threaded joints that connect two sides of the clamp. Tightening the screw allows to create the connection between the bodies, with the wanted position (of course in between the allowed boundaries), doing the reverse operation allows to disassociate the bodies to be adjusted / removed. Fasteners must belong to Class of Resistance 12.9 or below, Titanium-Alloy bolts are admitted as well.

1.4 Material Assignment

The material assigned to our group, named AlSi10Mg, is a common silicon-based aluminum alloy for additive manufacturing processes like SLM (Selective Laser Melting). This material is widely used for its balanced combination of lightness and mechanical strength. The behavior of the material under load varies through a wide range based on the post-processing heat and surface treatment applied to the component, as well as for the printing strategy. In our application, we decided to achieve a T6 treatment, as suggested in [2]. The bracket can then undergo micro-shot peening, so that we can improve the fatigue life and surface texture, as stated in [1].

Table 1.2: Mechanical Proprieties

Mechanical Property	Symbol	Value	Unit
Density	ρ	2670	[Kg/m ³]
Poisson's Ratio	ν	0,33	[-]
Young's Modulus	E	75000	[Mpa]
Bulk Modulus	K	74600	[Mpa]
Tangent Modulus	G	5000	[Mpa]
Tensile Yield Strenght	Sy	250	[Mpa]
Ultimate Tensile Strenght	Su (UTS)	400	[Mpa]
Fatigue Strenght	Sf	105	[Mpa]

2 Design Process

In this section the procedure of the design of the bracket will be explained, starting from the topology optimization up until the screw design and verification. The main used 3D CAD

software have been Spaceclaim, Siemens Solid Edge 2024 and PTC Creo Elements/Direct, the only FEM software used has been Ansys in the version 2024R2. The static and dynamic fatigue most important values have been retrieved by a convergence study, as shown in the “*Convergence Analysis Spreadsheet*” file.

2.1 Topology Optimization

To start the optimization process we created a raw workpiece using CAD software, so that we were able to fill the gaps between the limit surfaces and the analyzed pieces.

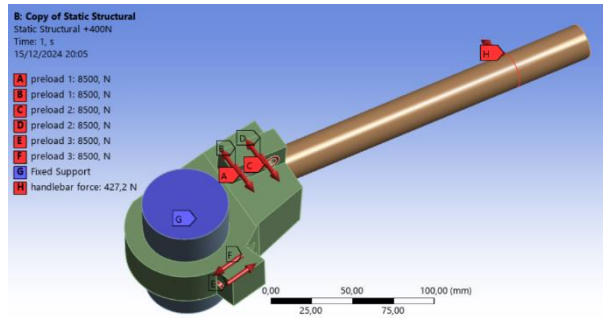


Figure 2.1 Topology Optimization setting

The settings of the topology optimization process are shown in the picture above, we split the handlebar surface at 50 mm from the end, so that we could apply the assigned load in that edge. It is important to state that the optimization was done considering two static structural, this has been done in order to consider loads in the “x” direction, both positive and negative, namely + 400 N and – 400 N. This will allow us to consider two static situations in which the driver could turn left, and therefore push the handlebar, as well as the other scenario of a right turn, so that the driver will drag the handlebar. Moreover, it is important to resist fatigue when the force is in both positive and negative directions.

We set a bonded contact in the internal ring of the clamp, and a bonded contact between the handlebar and the bracket, allowing the software to consider these as exclusion regions. To maintain the screw holes, (assumed to be M6 with analytical calculations, and then refined) we provided three cylindrical volumes of exclusion, in the underhead region. To avoid non linearities which are only supported in Ansys Structural Optimization by enabling the beta features, we applied two linear opposite forces instead of using the bolt pretension, we used linear contacts (all bonded) and linear “Aluminum Alloy” material for all the components. The forces applied to the screws are retrieved from the calculations on the screws, which will follow.

2.2 Topology geometry reconstruction

The topology optimization with a mass retain of 3% gave us a rough STL file that must be heavily reworked to get our final bracket. This process has been entirely covered inside Ansys Spaceclaim, which offers a large variety of tools for this aim. In particular, we had to round the edges of the four arms that connect the two clamps as much as possible, to prevent stress intensifications that could lead to failure, especially under fatigue with the assigned cyclic

load, as well as refining the internal cylindrical surfaces of the clamp, the top and bottom one of the stanchion clamp, the cuts of the clamps and the screws' hub, underhead and threaded holes. Moreover, to fully exploit the potential of additive technology, we also kept it as faithful as possible to the result of the topology optimization. This led us to a final model of the bracket ready for the validations explained in the following.

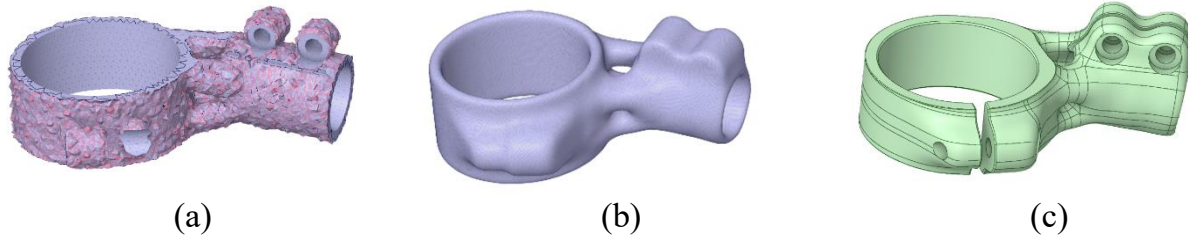


Figure 2.2: Ansys STL Output (a); Reconstruction with shrinkwrap (b); Final Bracket (c)

2.3 Screw Design and Verification

To select the size and the preload of the screws, it has been searched for similar components online and the choice was based on the admitted maximum preload by the VDI Table A1 trying on Ansys if the correspondent minimum preload was able to keep the parts connected. At the end three M6 - 8.8 socket head cap screws were selected with a minimum preload of 5 kN. This value is used in Ansys: it is enough to avoid the slipping of the clamp on the stanchion and to limit the stress generated by the handlebar. Finally, the screws were verified under maximum preload following the VDI 2230 approach.

2.2.1 Stanchion Clamp Screw Design

The clamp on the stanchion was modeled as a shaft-hub tightening couple following the Niemann approach [8] to determine the minimum screw preload required to prevent slipping (using a frictional coefficient of 0.2 on the surfaces stanchion – clamp and handlebar – clamp from the VDI table A6). The calculated minimum force was approximately 4 kN, incorporating a safety factor of 1.5 on the torque (detailed calculations can be found in “*Screw Calculations.pdf*”).

Then on VDI table A1 [9] the screw has been selected under the following assumptions:

- **Strength grade:** 8.8 to avoid thread stripping due to the lower strength of AlSi10Mg compared to the screw.
- **Thread friction coefficient:** 0.16 (class B, dry conditions, VDI Table A5)
- **Tightening factor (α_A):** 1.7 (torque wrench, VDI Table A8)

Based on the admitted maximum preload, M6 – 8.8 screw has been chosen (9.6 kN for utilization factor of 0.9). In this way a minimum preload up to 5.6 kN can be used ($F_{Mmin} = F_{Mtab}/\alpha_A$). A simulation with 5 kN of preload reached convergence, while a preload too low did not because of the slipping of the clamp.

M6 x 35 – 8.8 – ISO 4762.

2.2.2 Handlebar Clamp Screws

Moving to the handlebar clamp, 2 screws M6 - 8.8 are used to achieve more stability on the handlebar and to split the stress. We placed the screws on top because in this way the cut, which is the weaker part, is placed far away from the point where there is the highest contact pressure between the clamp and the handlebar (on the external edge of the clamp in the right and left bottom considering the alternate loads). The minimum preload was selected based on VDI table A1 of 5 kN under the same assumptions of the screw on the stanchion clamp. Also, in this case a simulation with this preload showed acceptable values of stress on the arms and small displacement of the clamp and handlebar, while a preload too low caused the opening of the handlebar's clamp and an increase of stress on the arms.

M6 x 20 – 8.8 – ISO 4762.

2.2.3 Screws Verification

The verification of the screws was completed following the VDI approach [9]: assembly stress, working stress, alternating stress, surface pressure, minimum length of engagement, tightening torque (see the file *screw_calculation.pdf*).

The assembly condition $F_{Mmax} < F_{Mtab}$ is verified because the screw was selected to stay under F_{Mtab} (assembly preload with 90% utilization of the minimum yield point).

Table 2.1: Preload of Screws

F_{Mmin} [N]	Minimum assembly preload	5000
F_{Mmax} [N]	Maximum assembly preload ($F_{Mmin} * \alpha_A$)	8500
F_{Mtab} [N]	Tabulated assembly preload ($v=0.9$)	9600

The total screw load in application F_{Smax} is obtained through the Ansys static analysis in the next chapter (working load on the screws at the final time, preloaded with F_{Mmax} at a previous time step). The following table shows the safety factors obtained from *screw_calculation.pdf*.

Table 2.2: Total Screw Loads and Design Parameters

	Screw 1	Screw 2	Screw 3
F _{Smax} [N]	8501.2	8860	8651.1
Working safety factor	1.35	1.30	1.33
Fatigue safety factor	1993.25	6.64	15.82
Surface pressure safety factor	1.98	1.90	1.94
Thread stripping safety factor	5.43	4.26	4.37

The fatigue safety factor of the first screw is remarkably high because the working load is very similar to the assembly preload, this could be caused by the fact that the other screws and the clamp's arms are able to withstand the stress without transmitting it to the screw. Another factor is that the clamp slips slightly around the stanchion, so the first screw isn't stretched too much so the load doesn't increase in exercise.

The minimum length of engagement is 7.4 [mm] ($1.25 \cdot d$) according to VDI figure 5.5/4, so it is used 9 mm and 11 mm for safety.

The tightening torque is obtained from the Motosh formula $M_A = 6590$ [N*mm].

3 Final Structure Validation

As stated before, also in the validation process of the bracket, convergence analysis data were used.

3.1 Static Analysis

To ensure that the clamp resists to the given loads we performed a static structural analysis on Ansys, to ensure that the maximum Von Mises stress stays below our admissible stress. We also performed a convergence validation to ensure that the discretized solution accurately approximates the real one and remains consistent over mesh size changes. We did this process in the areas where we found the highest stresses, and we found that the most stressed one, excluding the singularities in the screws' regions, was the one shown in the picture below.

The maximum Von Mises stress that we obtained was 117MPa that corresponds to a safety factor of 2.13 which is acceptable.

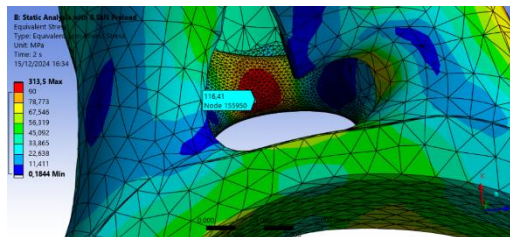


Figure 3.1: Static Structural Analysis

3.3 Handlebar Displacement

Our goal is to provide a bracket that can resist to the subjected stresses, but another important aspect to consider is that the biker must have a reliable steering, which could not be achieved if the handlebar has a high displacement whenever he steers.

For this purpose, we can consider our system as a cantilever beam where even a small rotation of the root of the bracket due to elastic deformation could lead to a high displacement at the other end of the bar, resulting in an uncomfortable steering. We checked this parameter in Ansys, which resulted in a maximum displacement of the handlebar of 2.14 mm which is acceptable under the static limit load, therefore approving the result.

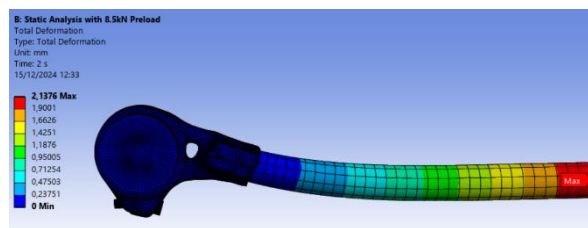


Figure 3.2: Total displacement of the handlebar

3.2 Fatigue Analysis

To achieve the fatigue verification of the bracket we had to set up calculations related to the life cycles and to the loads given for the purpose.

First of all, we searched for the S-N diagram of the AlSi10Mg, that we found in the literature, in the file [1]. Here we found the following S-N curve:

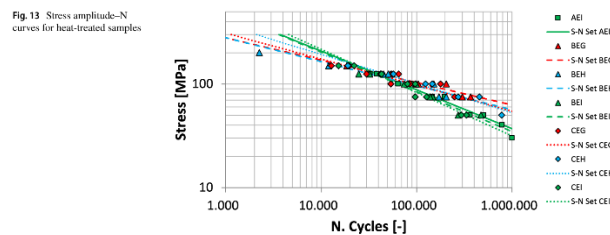


Figure 3.3: S-N Diagram of AlSi10Mg samples

From these different treated samples, we decided to take the “BEG” one, which according to the nomenclature, considered a 45° build orientation sample, treated with T6 process and followed by a micro-shot peening. Given the fact that the N value that we need to verify was 10^5 cycles, we took 105 MPa as Sf (stress fatigue).

From the same reference [1] we also took the values of Sy (yielding stress failure) and Sut (ultimate tensile stress), resulting in the numbers shown in Table 1.2.

We started from Ansys to find the weakest zone in terms of fatigue failure, which was the handlebar clamp one, including the four arms which are subjected to the highest stress amplitudes. In particular we tested different areas around of which there were the highest stresses and stress amplitudes, and we found out that the worst one is the one shown in the picture below, which didn't have the highest value of principal stress in the component and gave the lowest safety factor (2,09) after a convergence analysis to confirm the consistence of the result. This confirmed the importance of testing different areas of the component that had high stress amplitudes.

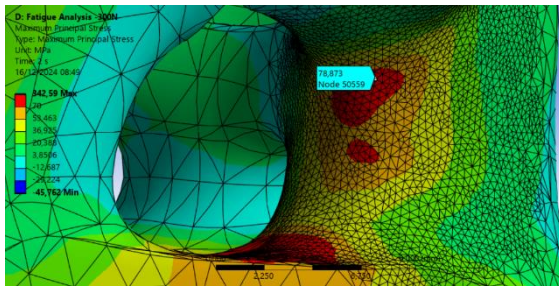


Figure 3.4: σ_{max}

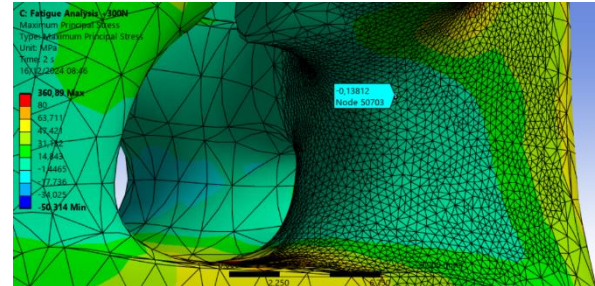


Figure 4.5: σ_{min}

We assumed, by reading ISO 12110-1, that a common safety factor (SF) for a high-end motorbike bracket should be between 1.2 and 2. Therefore, we made considerations about the fact that we are designing a component used in high performance motorbikes, and so the bracket must be as light as possible.

In this way, we found two range values corresponding to the maximum principal stresses in case of $F_x = \pm 300\text{N}$ that were accounted as σ_{max} and σ_{min} (shown in the picture above).

These values led us after all the detailed calculations present in the “*Convergence Analysis Spreadsheet*” file to a safety factor $SF = 2.09$.

3.3 Spatial boundaries

Once all the required verifications have been passed, we created the final assembly shown in the picture below. It can be noticed that it does not exceed the given constraints.

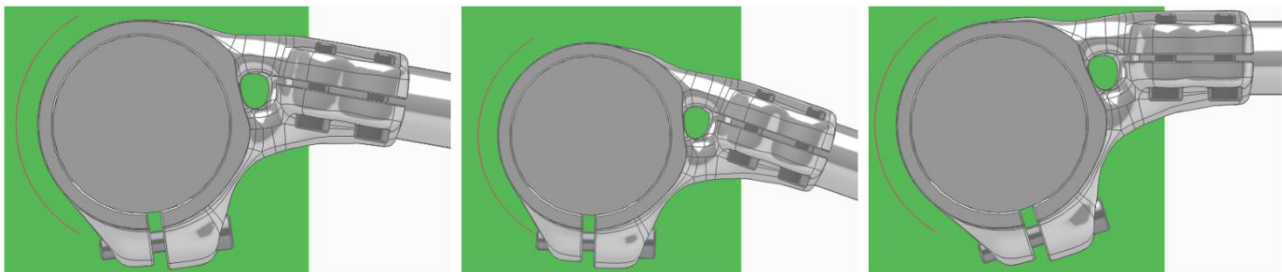


Figure 3.6: Three regulations (0° , 10° , -10°)

4 Manufacturing

4.1 Technical Drawing

Here it is the technical drawing of the component:

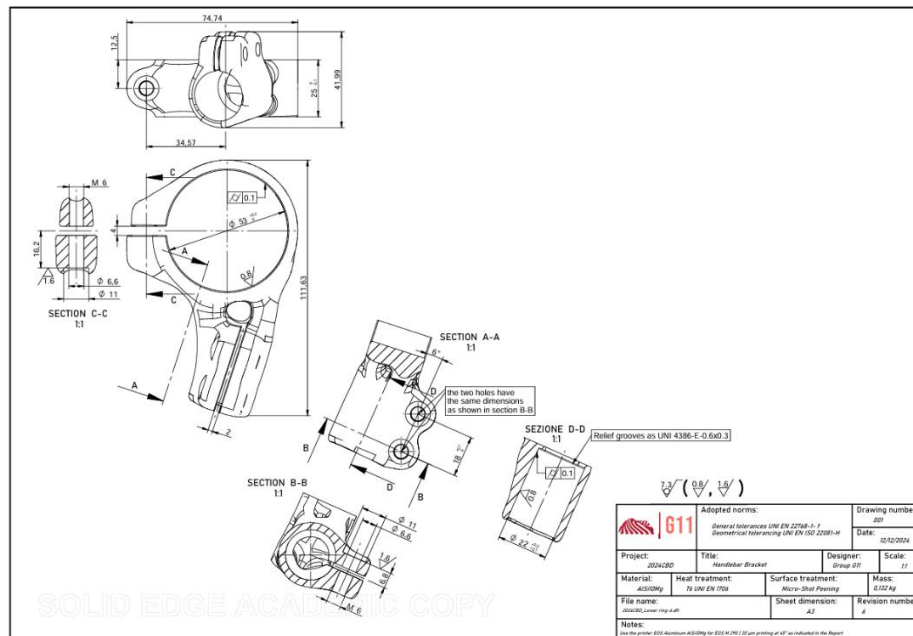


Figure 4.1: Technical Drawing

Since the part is firstly obtained with a non-machined manufacturing process, the surface texture and the mechanical properties are not suitable for our scopes, so the bracket is heat treated with a T6 treatment, and then micro-shot peened as surface treatment. We retrieved from the literature [1] that after the whole procedure the superficial texture is $Ra\ 7.3\ \mu m$. In order to make the bracket able to mate with the stanchion and the handlebar, it is necessary to machine the object in the mating areas, till a surface texture of $0.8\ \mu m$. It is important to also to guarantee the geometric mating between the surfaces, that is the reason behind the choice of the geometric tolerances of cylindricity. Furthermore, we set dimensional tolerances to have clearance fit, permitting the manual insertion, the handlebar is considered obtained by tube drawing so is basic shaft, while the stanchion is considered machined.

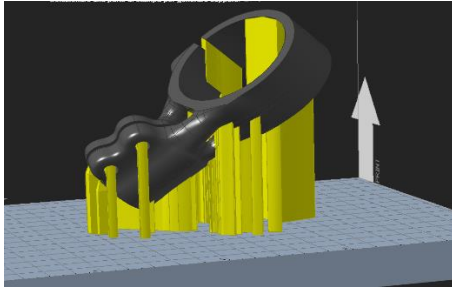
Of course, the mating surfaces between the bracket and the screws are also important, and so they will be machined till the surface has a reasonable Ra of $1.6\ \mu m$.

In the Notes section all the information concerning the manufacturing process of the represented object is indicated, leaving the printing and production processes to the next chapter.

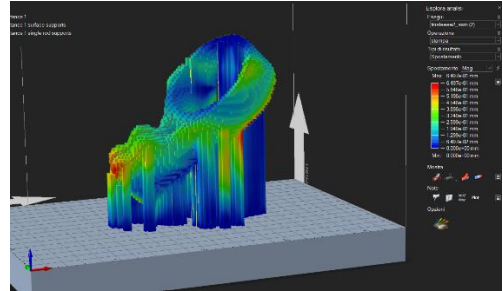
4.2 Printing Process

To conclude, we thought about a printing SLM process for our bracket. We have used Altair Inspire to achieve this analysis. The main part was to create printing supports in the correct

region of the environment, and to analyze their deformation due to the gradient of temperature over time.



(a)



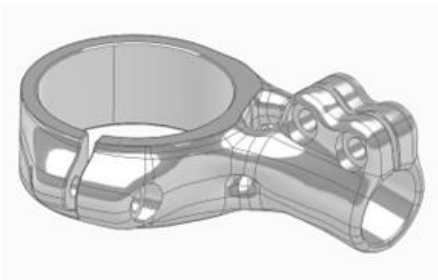
(b)

Figure 4.2: Printing Supports and Deformation Analysis

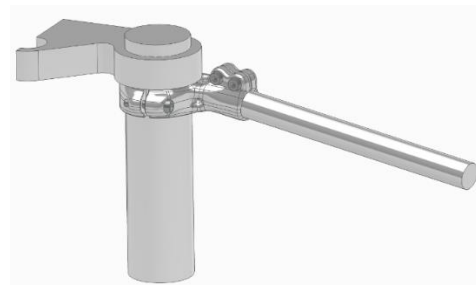
Using the thermal map on Altair, we found the best printing direction of the process, so that deformations were minimized at 45° , leaving a value of $\varepsilon_{max} = 0,65$ mm. This analysis is important because it gives an idea of the quantity of residual stresses that are present in the bracket. Minimizing the deformation will come with a compromise of printing time and support material. Given the fact that, with minimized deformations, the bracket will be printed in 4.5 hours and since is made for a high-end motorbike (so not a huge mass production), we approved the result given by the software as a printing solution. In the light of that, we selected “EOS Aluminum AlSi10Mg for EOS M 290 | $30\text{ }\mu\text{m}$ ” as printer, which will give shiny surfaces and good mechanical proprieties, as stated in [1].

Then T6 heat treatment and micro shot-peening are performed to improve mechanical properties. After that, we considered a 5 axis CNC milling machine, to refine the mating surfaces, the threaded holes, and the relief groove at the bottom of the handlebar clamp. Specific data for these processes can be seen in the technical drawing.

4.3 Final product and assembly



(a)



(b)

Figure 4.1: Final product(a); Final Assembly(b)

Bibliography

- [1] Fatigue response of AlSi10Mg by laser powder bed fusion: influence of build orientation, heat, and surface treatments. (S. Fini · D. Croccolo · M. De Agostinis · G. Olmi · L. Paiardini · C. Scapecchi · M. Mele)*
- [2] EOS Aluminum AlSi10Mg Material datasheet*
- [3] ETIMOS Aluminum datasheet*
- [4] ASM Aluminum 7000 series characterization*
- [5] Lezioni di fondamenti e tecnica della progettazione meccanica (D. Croccolo, N. Vincenzi)*
- [6] Esercizi di costruzioni di macchine e tecnica delle costruzioni meccaniche (D. Croccolo, M. De Agostinis, G. Olmi)*
- [7] Manuale di Meccanica- HOEPLI (L. Caligaris, S. Fava, C. Tomasello)*
- [8] Machine elements design and calculation in mechanical engineering: vol.1 (G. Niemann)*
- [9] VDI 22230 Systematic calculation of high duty bolted joints. Joints with one cylindrical bolt*